

Section : Chemistry

Q.1 The following statements describe various properties of a Mercury cell:

- (A) It converts energy of combustion into electrical energy
- (B) It is rechargeable
- (C) The cell reaction involved is $\text{Zn(Hg)} + \text{HgO(s)} \rightarrow \text{ZnO(s)} + \text{Hg(l)}$
- (D) It is a low current device used in hearing aids

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.2 Hell-Volhard Zelinsky reaction is used for the formation of

- 1. alcohols
- 2. aldehydes
- 3. ketones
- 4. α -halocarboxylic acids

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.3 Which of the following does not represent a correct application of the coordination compound?

- 1. *cis*-platin effectively inhibit the growth of tumours.
- 2. chlorophyll help in photosynthesis
- 3. desferrioxime B is used in treatment of lead poisoning
- 4. cyanocobalamine, antipernicious anemia factor is a coordination compound of Co

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.4 What is the decreasing order of field strength of given ligands?

- (A) S^{2-}
- (B) Ethylenediamine
- (C) NCS^-
- (D) CN^-

Choose the correct answer from the options given below:

- 1. (A), (C), (B), (D)
- 2. (A), (B), (C), (D)
- 3. (D), (B), (C), (A)
- 4. (D), (C), (B), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.5 What is an example of globular protein?

- (A) Insulin
- (B) Keratin
- (C) Albumin
- (D) Myosin

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.6 What is the correct sequence of increasing reactivity of the following compounds towards nucleophilic addition reaction?

- (A) Ethanal
- (B) Propanone
- (C) Propanal
- (D) Butanone

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (D), (B), (C), (A)
- 3. (A), (C), (B), (D)
- 4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Acidified potassium dichromate oxidizes sulphides (S^{2-}) to

1. SO_4^{2-}

2. SO_3^{2-}

3. sulphur (S)

4. SO_2

Options 1. 1

2. 2

3. 3

4. 4

Q.8 Gabriel phthalimide synthesis is used for the preparation of

- 1. tertiary amine
- 2. acid synthesis
- 3. primary amine
- 4. secondary amine

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Select the correct statements for $[Fe(CN)_6]^{3-}$ complex:

(A) Paramagnetic

(B) sp^3d^2 hybridization

(C) Magnetic moment = 5.92 BM

(D) d^2sp^3 hybridization

Choose the correct answer from the options given below:

- 1. (A) and (D) only
- 2. (A), (B) and (C) only
- 3. (B) and (C) only
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.10 The value of van't Hoff factor, i , for CH_3COOH solution in water will be

- 1. Between 1 and 2
- 2. Less than 1
- 3. 2
- 4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Arrange the following compounds in increasing order of their acidic strength:

- (A) 3-nitrophenol
- (B) 3,5-Dinitrophenol
- (C) 2,4,6 -Trinitrophenol
- (D) Phenol

Choose the correct answer from the options given below

1. (D), (C), (B), (A)
2. (C), (A), (B), (D)
3. (D), (A), (B), (C)
4. (A), (B), (C), (D)

Options

1. 1
2. 2
3. 3
4. 4

Q.12 Calculate the \text{molality} of KI if the density of 20% (mass/mass) aqueous solution of KI is 1.202 g mL^{-1} .
(Molar mass of KI is 166 g mol^{-1})

1. 1.5 mol kg^{-1}
2. 1.2 mol kg^{-1}
3. 15 mol kg^{-1}
4. 0.12 mol kg^{-1}

Options

1. 1
2. 2
3. 3
4. 4

Q.13 The unit of E_{cell} is

1. V m^{-1}
2. S cm^{-1}
3. V
4. $\text{S cm}^{-2} \text{ mol}^{-1}$

Options

1. 1
2. 2
3. 3
4. 4

Q.14 The structural feature in carbonyl compound for Aldol condensation

1. presence of at least one β -hydrogen
2. presence of at least one α -hydrogen
3. Concentrated base
4. lack of α hydrogen

Options

1. 1
2. 2
3. 3
4. 4

Q.15 A galvanic cell behaves as electrolytic cell when ?

1. $E_{\text{cell}} = E_{\text{ext}}$
2. $E_{\text{cell}} = 0$
3. $E_{\text{ext}} > E_{\text{cell}}$
4. $E_{\text{cell}} > E_{\text{ext}}$

Options

1. 1
2. 2
3. 3
4. 4

Q.16 Which of the following compounds will not give azo coupling reaction with benzene diazonium chloride?

1. Nitrobenzene
2. Aniline
3. o-Toluidine
4. Phenol

Options

1. 1
2. 2
3. 3
4. 4

Q.17 Why is HCl not used to make the medium acidic in oxidation reactions of KMnO_4 in acidic medium?

1. KMnO_4 is weaker oxidizing agent than HCl
2. KMnO_4 oxidises HCl into Cl_2 which is also an oxidizing agent
3. Both HCl and KMnO_4 act as oxidizing agent
4. KMnO_4 act as a reducing agent in the presence of HCl

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Optically active alkyl halide undergoing $\text{S}_\text{N}2$ substitution involves

1. retention of configuration
2. racemic mixture
3. inversion of configuration
4. formation of carbocation

Options 1. 1

2. 2
3. 3
4. 4

Q.19 The following solutions were prepared by dissolving 1 g of solute in 1 L of the solution. Arrange the following solutions in decreasing order of their molarity

- (A) Glucose (molar mass = 180 g mol^{-1})
(B) NaOH (molar mass = 40 g mol^{-1})
(C) NaCl (molar mass = 58.5 g mol^{-1})
(D) KCl (molar mass = 74.5 g mol^{-1})

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (A), (B), (D), (C)
3. (B), (C), (D), (A)
4. (D), (C), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Arrange the following ions in increasing order of number of 3d-electrons

- (A) Cr^{2+}
(B) Cu^+
(C) Ti^{3+}
(D) Mn^+

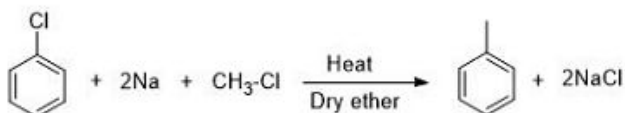
Choose the correct answer from the options given below:

1. (B), (A), (C), (D)
2. (C), (A), (D), (B)
3. (C), (D), (A), (B)
4. (D), (B), (C), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.21



The above reaction is an example of

1. Sandmeyer's reaction
2. Wurtz reaction
3. Wurtz Fittig reaction
4. Kolbe reaction

Options

1. 1
2. 2
3. 3
4. 4

Q.22 Which reagent will distinguish Benzophenone from acetone ?

1. Fehling's reagent
2. Tollen's reagent
3. 2,4-DNP reagent
4. I₂/NaOH

Options

1. 1
2. 2
3. 3
4. 4

Q.23 Which types of bonds or interactions are found in the β -helix of protein?

1. Ionic bond
2. Covalent interaction
3. H-bond
4. Banana bond

Options

1. 1
2. 2
3. 3
4. 4

Q.24 The atomic number of Lanthanum is 57. Its electronic configuration will be

1. [Xe]5d¹6s²
2. [Xe]4f¹5d²
3. [Xe]4f³
4. [Xe]4f¹5d¹6s¹

Options

1. 1
2. 2
3. 3
4. 4

Q.25 Give the reason for low concentration of oxygen in the blood and tissues of people living at high altitude

1. Both low temperature and high atmospheric pressure
2. low temperature
3. Low atmospheric pressure
4. High atmospheric pressure

Options

1. 1
2. 2
3. 3
4. 4

Q.26 What is the IUPAC name of [Pt(NH₃)₂Cl₂] ?

1. Diamminedichloridoplatinum (IV)
2. Diamminedichloridoplatinum (II)
3. Diamminedichloridoplatinum (0)
4. Dichlorodiammineplatinum (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Match the amino acid given in List-I with their one letter code given in List-II

List-I	List-II
Name of amino acid	One letter code
(A) Lysine	(I) W
(B) Tryptophan	(II) Q
(C) Tyrosine	(III) K
(D) Glutamine	(IV) Y

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Match compound/elements of List-I with their uses given in List-II

List-I	List-II
(Compound/Elements)	(Uses)
(A) Magnesium based alloy is constituent of	(I) Bullets
(B) Lanthanoid oxide	(II) Petroleum cracking
(C) Mixed oxides of Lanthanoids are employed in	(III) Television screen
(D) Misch metal	(IV) Lanthanoid metal and iron

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Match List-I with List-II

List-I	List-II
(A) Ambident nucleophiles	(I) Symmetrical object
(B) Plane polarized light	(II) Saytzeff rule
(C) Superimposable mirror image	(III) Cyanides and nitrites
(D) β -elimination reaction	(IV) Nicol Prism

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.30 The nitrogen atom in amines is trivalent and possess an unshared pair of electrons. The geometry of trimethyl amine is

1. Tetrahedral
2. Pyramidal
3. Square planar
4. Triangular

Options 1. 1

2. 2

3. 3

4. 4

Q.31 The carbohydrate used as storage molecule in plants is

1. Starch
2. Glycogen
3. Cellulose
4. Glucose

Options 1. 1

2. 2

3. 3

4. 4

Q.32 What is the color of $\text{Fe}^{3+}(\text{aq})$ ion?

1. Yellow
2. Violet
3. Colourless
4. Bluepink

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Match List-I with List-II

List-I

List-II

Solutions

Explanation

(A) Saturated solution

(I) Solution having two components.

(B) Isotonic solutions

(II) A solution whose osmotic pressure is more than that of another.

(C) Binary solution

(III) A solution which contains the maximum amount of solute that can be dissolved in a given amount of solvent at a given temperature.

(D) Hypertonic solution

(IV) The solutions having same osmotic pressure at a given temperature.

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 The reagent(s) used in hydroboration oxidation of propene are

- (A) B_2H_6
- (B) H_2O
- (C) H_2O_2
- (D) OH^-

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Match List-I with List-II

List-I

List-II

Property

Unit

(A) Cell constant

(I) cm^{-1}

(B) Molar conductance (II) $\text{ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$

(C) Specific conductance (III) $\text{ohm}^{-1} \text{cm}^{-1}$

(D) Conductance (IV) ohm^{-1}

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.36 What happens when $\text{C}_6\text{H}_5\text{-O-R}$ is treated with HX ?

1. RX and $\text{C}_6\text{H}_5\text{OH}$ are formed

2. ROH and $\text{C}_6\text{H}_5\text{X}$ are formed

3. $\text{C}_6\text{H}_4\text{X}_2$ and ROH are formed

4. RX and $\text{C}_6\text{H}_5\text{X}$ are formed

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Which cell is used in automobiles and inverters?

1. Mercury cell

2. Dry cell

3. Lead storage cell

4. Fuel cell

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Which of the following reagent(s) is required for the conversion of Benzene to methyl benzoate?

(A) $\text{Br}_2/\text{FeBr}_3$

(B) Mg , dry Ether

(C) CO_2 , H_3O^+

(D) Methanol, Conc. H_2SO_4

Choose the correct answer from the options given below:

1. (A), (B) and (C) only

2. (A), (B) and (D) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.39 In the \textbf{nitration of benzene} using a mixture of conc. H_2SO_4 and conc. HNO_3 , the nitrating species is

1. NO_2^-

2. NO_2^+

3. NO^+

4. NO_2 and NO_2^+

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Amylose is a water-soluble part of starch. What is the percentage solubility of it?

1. 20 to 30%

2. 15 to 20%

3. 30 to 60%

4. 50 to 70%

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the questions

The speed at which a chemical reaction takes place is called the rate of reaction. The rate of reaction depends on various factors like concentration of the reactants, temperature, etc. The relation between the rate of reaction and the concentration of reacting species is represented by the equation $r = k[A]^x[B]^y$, where x and y are the order of the reaction with respect to the reactants A and B, respectively. The overall order of the reaction is $x + y$. The rate of reaction can also be increased by the use of a catalyst which provides an alternate pathway of lower activation energy. It increases the rate of forward and backward reaction to an equal extent. It does not alter the Gibbs energy of the reaction.

The rate of a gaseous reaction is given by $r = k[A][B]$. If the volume of the reaction vessel is suddenly reduced to $(1/4)$ th of its initial value, the reaction rate relating to the initial rate will become

1. 2 times
2. $1/4$ times
3. 16 times
4. $1/16$ times

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.42 Read the passage carefully and answer the questions

The speed at which a chemical reaction takes place is called the rate of reaction. The rate of reaction depends on various factors like concentration of the reactants, temperature, etc. The relation between the rate of reaction and the concentration of reacting species is represented by the equation $r = k[A]^x[B]^y$, where x and y are the order of the reaction with respect to the reactants A and B, respectively. The overall order of the reaction is $x + y$. The rate of reaction can also be increased by the use of a catalyst which provides an alternate pathway of lower activation energy. It increases the rate of forward and backward reaction to an equal extent. It does not alter the Gibbs energy of the reaction.

Calculate the order of a reaction whose Rate $= k[A]^{1/2}[B]^{3/2}$.

1. second order
2. half order
3. first order
4. zero order

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.43 Read the passage carefully and answer the questions

The speed at which a chemical reaction takes place is called the rate of reaction. The rate of reaction depends on various factors like concentration of the reactants, temperature, etc. The relation between the rate of reaction and the concentration of reacting species is represented by the equation $r = k[A]^x[B]^y$, where x and y are the order of the reaction with respect to the reactants A and B, respectively. The overall order of the reaction is $x + y$. The rate of reaction can also be increased by the use of a catalyst which provides an alternate pathway of lower activation energy. It increases the rate of forward and backward reaction to an equal extent. It does not alter the Gibbs energy of the reaction.

If the rate of reaction becomes twenty-seven times upon increasing the concentration of reactant by three times, the order of this reaction is

- 1.0
- 2.1
- 3.3
- 4.2

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.44 Read the passage carefully and answer the questions

The speed at which a chemical reaction takes place is called the rate of reaction. The rate of reaction depends on various factors like concentration of the reactants, temperature, etc. The relation between the rate of reaction and the concentration of reacting species is represented by the equation $r = k[A]^x[B]^y$, where x and y are the order of the reaction with respect to the reactants A and B, respectively. The overall order of the reaction is $x + y$. The rate of reaction can also be increased by the use of a catalyst which provides an alternate pathway of lower activation energy. It increases the rate of forward and backward reaction to an equal extent. It does not alter the Gibbs energy of the reaction.

The rate law of a reaction is given by $r = k[\text{CH}_3\text{OCH}_3]^{3/2}$. If the pressure is measured in bar and time in minutes, then the unit of rate constant will be

1. $\text{bar}^{1/2} \text{ min}^{-1}$.
2. $\text{bar}^{-3/2} \text{ min}^{-1}$.
3. $\text{bar}^{-1/2} \text{ min}^{-1}$.
4. $\text{bar}^{3/2} \text{ min}^{-1}$.

Options

1. 1
2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the questions

The speed at which a chemical reaction takes place is called the rate of reaction. The rate of reaction depends on various factors like concentration of the reactants, temperature, etc. The relation between the rate of reaction and the concentration of reacting species is represented by the equation $r = k[A]^x[B]^y$, where x and y are the order of the reaction with respect to the reactants A and B, respectively. The overall order of the reaction is $x + y$. The rate of reaction can also be increased by the use of a catalyst which provides an alternate pathway of lower activation energy. It increases the rate of forward and backward reaction to an equal extent. It does not alter the Gibbs energy of the reaction.

The role of a catalyst is to change

1. Gibbs energy of the reaction.
2. Enthalpy of a reaction.
3. Activation energy of a reaction.
4. Equilibrium constant.

Options

1. 1
2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the questions

Replacement of a hydrogen atom in a hydrocarbon by an alkoxy or carboxyl group yields a class of compounds known as ethers. Ethers are classified as symmetrical or unsymmetrical on the basis of groups attached to the oxygen atoms. Diethyl ether, a symmetrical ether, has been widely used as an inhalation anesthetic. Ethers can be prepared by acid catalyzed intermolecular dehydration of alcohols and Williamson's synthesis. Acid catalyzed dehydration of alcohols is not generally preferred as it gives a mixture of elimination and substitution products. In Williamson's synthesis, an alkyl halide is allowed to react with sodium alkoxide. Ethers containing substituted Alkyl groups may also be prepared by this method. The C-O bond in ether is weakly polar and is cleaved under drastic conditions with excess of hydrogen halides. In electrophilic substitution, the alkoxy group deactivates the aromatic ring and directs the incoming group to ortho and para positions.

In \textbf{Williamson synthesis}, the alkoxide ion attacks the alkyl halide via which pathway?

1. $\text{S}_{\text{N}}2$
2. $\text{S}_{\text{N}}1$
3. Depends on nature of alkoxide ion
4. Depends on the nature of Alkyl halide

Options

1. 1
2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the questions

Replacement of a hydrogen atom in a hydrocarbon by an alkoxy or carboxyl group yields a class of compounds known as

ethers. Ethers are classified as symmetrical or unsymmetrical on the basis of groups attached to the oxygen atoms. Diethyl ether, a symmetrical ether, has been widely used as an inhalation anesthetic. Ethers can be prepared by acid catalyzed intermolecular dehydration of alcohols and Williamson's synthesis. Acid catalyzed dehydration of alcohols is not generally preferred as it gives a mixture of elimination and substitution products. In Williamson's synthesis, an alkyl halide is allowed to react with sodium alkoxide. Ethers containing substituted Alkyl groups may also be prepared by this method. The C-O bond in ether is weakly polar and is cleaved under drastic conditions with excess of hydrogen halides. In electrophilic substitution, the alkoxy group deactivates the aromatic ring and directs the incoming group to ortho and para positions.

Which type of ether is anisole ?

1. Dialkyl ether
2. Diaryl ether
3. Phenyl Alkyl ether
4. Alkoxy Alkyl ether

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions

Replacement of a hydrogen atom in a hydrocarbon by an alkoxy or carboxyl group yields a class of compounds known as ethers. Ethers are classified as symmetrical or unsymmetrical on the basis of groups attached to the oxygen atoms. Diethyl ether, a symmetrical ether, has been widely used as an inhalation anesthetic. Ethers can be prepared by acid catalyzed intermolecular dehydration of alcohols and Williamson's synthesis. Acid catalyzed dehydration of alcohols is not generally preferred as it gives a mixture of elimination and substitution products. In Williamson's synthesis, an alkyl halide is allowed to react with sodium alkoxide. Ethers containing substituted Alkyl groups may also be prepared by this method. The C-O bond in ether is weakly polar and is cleaved under drastic conditions with excess of hydrogen halides. In electrophilic substitution, the alkoxy group deactivates the aromatic ring and directs the incoming group to ortho and para positions.

When ethanol is dehydrated in the presence of H_2SO_4 , at 443 K and 413 K respectively, the products formed are

1. Ethane and ethoxythane
2. Ethylmethyl ether and butene
3. Ethylmethyl ether and propene
4. Ethene and ethoxyethane

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions

Replacement of a hydrogen atom in a hydrocarbon by an alkoxy or carboxyl group yields a class of compounds known as ethers. Ethers are classified as symmetrical or unsymmetrical on the basis of groups attached to the oxygen atoms. Diethyl ether, a symmetrical ether, has been widely used as an inhalation anesthetic. Ethers can be prepared by acid catalyzed intermolecular dehydration of alcohols and Williamson's synthesis. Acid catalyzed dehydration of alcohols is not generally preferred as it gives a mixture of elimination and substitution products. In Williamson's synthesis, an alkyl halide is allowed to react with sodium alkoxide. Ethers containing substituted Alkyl groups may also be prepared by this method. The C-O bond in ether is weakly polar and is cleaved under drastic conditions with excess of hydrogen halides. In electrophilic substitution, the alkoxy group deactivates the aromatic ring and directs the incoming group to ortho and para positions.

The major product in the reaction of anisole with bromine in ethanoic acid, is:

1. o- bromoanisole
2. p- bromoanisole
3. m-bromoanisole
4. o-bromoanisole and p-bromoanisole

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions

Replacement of a hydrogen atom in a hydrocarbon by an alkoxy or carboxyl group yields a class of compounds known as ethers. Ethers are classified as symmetrical or unsymmetrical on the basis of groups attached to the oxygen atoms. Diethyl ether, a symmetrical ether, has been widely used as an inhalation anesthetic. Ethers can be prepared by acid catalyzed intermolecular dehydration of alcohols and Williamson's synthesis. Acid catalyzed dehydration of alcohols is not generally preferred as it gives a mixture of elimination and substitution products. In Williamson's synthesis, an alkyl halide is allowed to react with sodium

alkoxide. Ethers containing substituted Alkyl groups may also be prepared by this method. The C-O bond in ether is weakly polar and is cleaved under drastic conditions with excess of hydrogen halides. In electrophilic substitution, the alkoxy group deactivates the aromatic ring and directs the incoming group to ortho and para positions.

Which is most reactive hydrogen halide for cleavage of ethers?

1. HF
2. HCl
3. HBr
4. HI

Options

1. 1
2. 2
3. 3
4. 4